

	Case Name: Pumping Station Jabbeke	Sector	Construction (civil)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Niels Crommen		User defined distributions
Date	2012	Project Control	Automatic tracking
File Name	C2012-13 Pumping Station Jabbeke		Tracking based on user input

1. Project description

Project authenticity

The construction and installation of a pumping station in Jabbeke (Belgium), including mechanical structures, equipment placement, and supporting infrastructure. The project consists of activity, resource, and cost data obtained from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	75	Serial/Parallel (SP)	64%
Planned Duration (PD)	125d	Activity Distribution (AD)	59%
Budget At Completion (BAC)	336,410 €	Length of Arcs (LA)	3%
Renewable Resources	2	Topological Float (TF)	27%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11	10	3.13
CRI-rho	11	10	3.13
CRI-tau	10	80	1.97

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	13	10	8.6
CRI-rho	13	10	0.1
CRI-tau	0	10	8.6

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	13	46	-0.4
SI	66	40	-0.5
SSI	4	10	6.1
CRI-r	10	11	4.7
CRI-rho	11	11	4.42
CRI-tau	10	10	3.40

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	14	-14
PV - SPI	21	4
PV - SCI	21	4
ED - 1	21	-21
ED - SPI	21	4
ED - SCI	21	4
ES - 1	12	-11
ES - SPI(t)	19	15
ES - SCI(t)	19	15

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0	0
CPI	0	0
SPI	12	12
SPI(t)	15	15
SCI	12	12
SCI(t)	15	15
0.8 CPI + 0.2 SPI	4	4
0.8 CPI + 0.2 SPI(t)	6	5

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **28** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-5716.75	-6470.76	-6	0.980	1.00	0.97	0.98
std dev	5479.76	18080.92	9.05	0.019	0.10	0.12	0.04
final	-14,101 €	0	-1.88	0.96	1.00	0.89	1

2.3.3.2. Time forecasting

PD	125d	Real Duration	140d	Late/early	12%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	146.97	9.41	6.42	-4.98
PV - SPI	144.43	15.98	9.49	-3.17
PV - SCI	148.55	19.14	12.78	-6.10
ED - 1	148.40	10.56	7.28	-6.00
ED - SPI	145.99	17.20	10.60	-4.28
ED - SCI	147.03	17.88	11.63	-5.02
ES - 1	151.53	12.90	9.28	-8.24
ES - SPI(t)	150.76	21.48	14.91	-7.69
ES - SCI(t)	151.88	22.53	16.06	-8.48

2.3.3.3. Cost forecasting

BAC	336,410 €	Real Cost	350,511.31€	Over/under budget	4%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	343265.59	6753.62	1.52	1.00
CPI	337584.84	18365.36	1.63	0.67
SPI	338318.77	21425.35	3.83	2.31
SPI(t)	338859.76	19955.35	4.79	2.10
SCI	339645.73	22888.72	4.12	1.94
SCI(t)	341965.85	9190.97	5.21	1.71
0.8 CPI + 0.2 SPI	342046.06	9636.49	2.10	1.04
0.8 CPI + 0.2 SPI(t)	343265.59	6753.62	2.29	1.02